ECON 164: Theory of Economic Growth

Week 5: AK Models

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Spring 2026

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- Next few lectures: models of "endogenous growth"

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- anchor to Solow, but everything we say applies to Ramsey-Cass-Koopmans too

The AK model: A limiting case of the Solow model

What happens in the **Solow model**. . . (with *constant* Hicks-neutral productivity)

$$\boldsymbol{Y_t = AK_t^{\alpha}(L_t)^{1-\alpha}}, \qquad \boldsymbol{\dot{K}_t = sY_t - \delta K_t}$$

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$$Y_t = AK_t, \qquad \dot{K}_t = sY_t - \delta K_t$$

...so the growth rate of capital becomes

$$g_K = \frac{\dot{K}_t}{K_t} = sA - \delta$$

(Problem Set #2, Question 1B)

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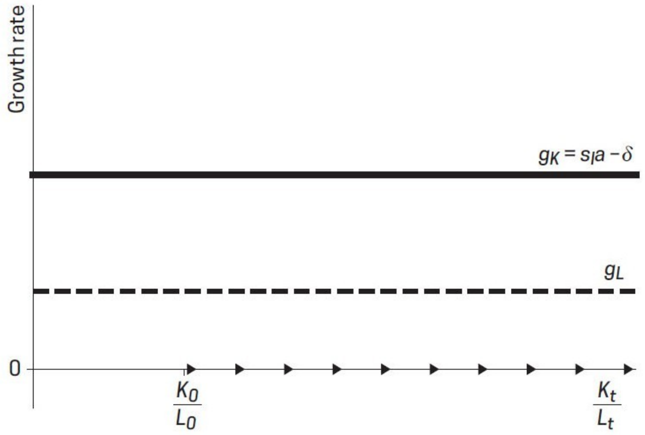
Figure 11.1 Dynamics of the *AK* Model



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Some observations:

- $\boldsymbol{g_y}$ is **independent** of k_t
- $\boldsymbol{g_y}$ is **positive** and **persistent** even w/ $g_A = 0$
- $\boldsymbol{g_y}$ is affected **permanently** by a change in policy ($\boldsymbol{s}$)

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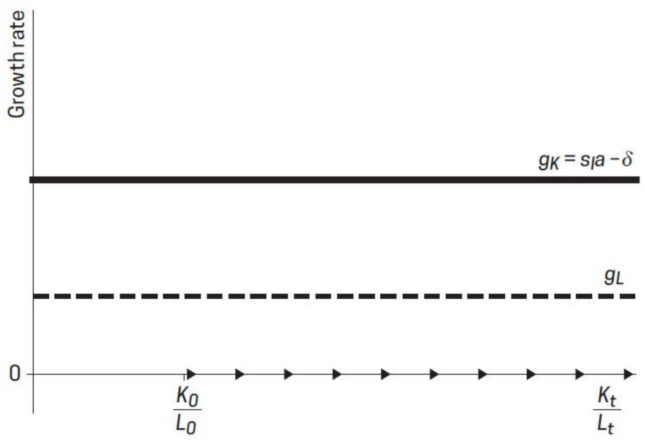

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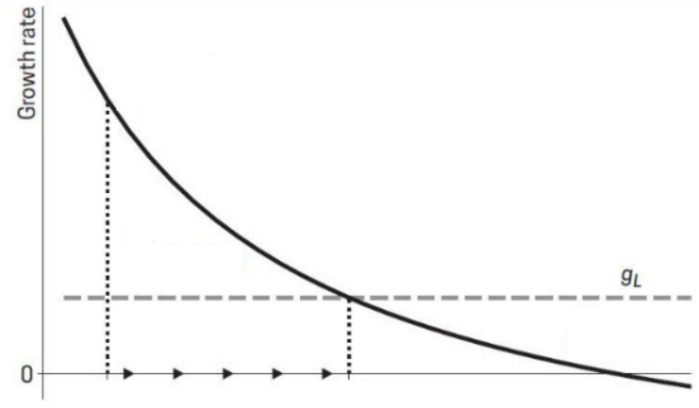

For $\alpha < 1$ ...

$$\dot{k} = s\mathbf{A}k^{\alpha} - (\delta + g_L)k$$

$$\downarrow$$

$$g_y = \alpha(sAk^{\alpha-1} - \delta - g_L)$$

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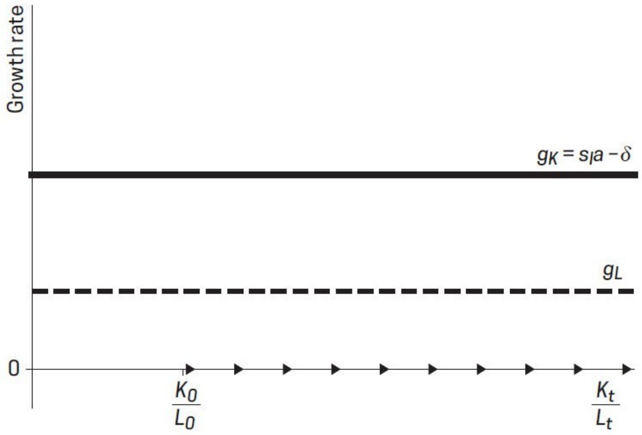

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. . . but with $\alpha = 1$:

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In **general**, there must be some variable X_t governed by a **linear differential equation**

$$\dot{X}_t = g_X X_t$$

(*why isn't* $\dot{L}_t = g_L L_t$ *enough then?*)

- Recall the **production function** with human capital (but *constant* Hicks-neutral A)

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where we introduce $\boldsymbol{u} \equiv$ **exogenous share** of labor used for production

The Uzawa-Lucas model

(Uzawa, 1965; Lucas, 1988)

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- Accumulating **physical capital** happens as it does in Solow...

$$\dot{K}_t = sY_t - \delta K_t$$

- ...but now we explicitly accumulate **human capital**, too:

$$\dot{h}_t = (1-u)h_t$$

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On the one hand, this is a cousin of the **AK model** $\rightarrow$ **same long-run dynamics**

- $\dot{\boldsymbol{h}}_t$ is linear in $\boldsymbol{h}_t$
- $\dot{\boldsymbol{K}}_t$ is linear in $\{\boldsymbol{K}_t, \boldsymbol{h}_t\}$ because $\boldsymbol{Y}_t = \boldsymbol{F}(\boldsymbol{K}_t, \boldsymbol{h}_t, \boldsymbol{L}_t)$ is **CRS** in that tuple

$$\begin{aligned}
\dot{\boldsymbol{K}}_t(\boldsymbol{\lambda K}_t, \boldsymbol{\lambda h}_t) &= sF(\lambda K_t, \lambda h_t, L_t) - \delta\lambda K_t \\
&= s\lambda F(K_t, h_t, L_t) - \delta\lambda K_t = \boldsymbol{\lambda}\dot{\boldsymbol{K}}_t(\boldsymbol{K}_t, \boldsymbol{h}_t)
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On the other, it's still a cousin of the **Solow model** → same transitional dynamics

- h_t acts like **labor-augmenting** productivity
- balanced growth path (BGP) requires **constant** $\hat{k}_t = \frac{K_t}{h_t L_t} \ldots$
- …so both s and u affect the level, but only u affects the growth rate of y_t

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- This maintains assumption of **perfect competition** $\rightarrow$ vs. next week

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2. scale effects
3. what do firms actually do?

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[advanced reading: McGrattan (1998)]

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- . . . but maybe we just need to think harder about the right **spatial scale**:
  - maybe it's the **whole world** in the very long run (Kremer, 1993)
  - maybe it's just **local labor markets**

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That's what we cover the next two weeks!

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when only transitory, prefer to call this ***semi-endogenous***

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References

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